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DEEP LEARNING MINI PROJECT REPORT

INTRODUCTION

Deep learning is a branch of machine learning that uses artificial neural networks inspired by the human brain. Instead of programmers manually designing rules, deep learning models *learn patterns automatically* from large amounts of data.

These models are built from multiple layers of neurons that process information step-by-step — which is why it's called *"deep."* As data passes through the layers, the model gradually learns features, such as edges in images, shapes, objects, or even meaning in language.

Deep learning powers many technologies we use every day, including image recognition, speech assistants, translation systems, medical diagnosis tools, and self-driving cars. Its strength comes from handling complex, high-dimensional data and improving as more data becomes available

The Cotton Plant is a shrub from the family of Malvaceae. It grows on every continent, as there are many varieties suited to a range of climates, and many ways of growing it. Cotton plants are cultivated for their fruits, known as "bolls". Once ripe, the bolls split to reveal a small white ball: cotton. The raw product is known as seed cotton, which is ginned to separate the fibre from the seeds. The fibre is spun to make yarn and woven to make fabric. The seeds provide oil and press cake.

Cotton, Gossypium, is a shrub from the family Malvaceae, which includes hollyhocks, mallow, hibiscus, okra, and cocoa. The genus Gossypium includes 50 cotton species, four of which have been domesticated for the fibre produced by their seeds. Two species that originated in the Americas now account for most of the cotton produced worldwide: Gossypium hirsutum, from Mexico, which produces some 90%, and Gossypium barbadense, from Barbados, which produces the best fibre but only accounts for around 5% of the total. The other two species originated in the Old World: Gossypium herbaceum, from southern Africa, and Gossypium arboreum, from India, account for around 5% of global output. Their shorter, thicker fibres are generally used by local spinners and weavers. The high importance of this crop has led to the need for proper classification of cotton plants with respect to different criteria using optimal models for better understanding and improved production.

Contribution of Deep Learning to Agricultural Development

Deep learning has become a powerful tool in modern agriculture**.** By analyzing images, sensor data, and satellite information, deep learning models help farmers monitor crops, detect plant diseases early, identify pests, and estimate yields with high accuracy. These systems can automatically recognize

subtle patterns—such as leaf discoloration or growth stress—that are difficult for the human eye to detect.

Deep learning also supports precision farming, where resources like water, fertilizer, and pesticides are applied only where needed. This reduces costs, minimizes environmental impact, and improves productivity. In addition, automated drones, robots, and smart irrigation systems rely on deep learning to make real-time decisions in the field.

Overall, deep learning contributes to higher crop quality**,** increased efficiency, and sustainable agricultural practices, helping farmers meet growing global food demands.

OBJECTIVES

The objective of this study is to investigate the performance of multiple Convolutional Neural Network (CNN) architectures on the Cotton_Disease dataset and determine the most effective model for classifying images into four categories: diseased cotton leaf, diseased cotton plant, fresh cotton leaf, and fresh cotton plant. Model performance will be evaluated by comparing predicted labels with ground-truth annotations and analyzing both training and test accuracies. Furthermore, the study aims to interpret the decision-making process of the best-performing model in order to identify the visual features and regions of the cotton plant that most strongly influence classification outcomes using Explainable Artificial Intelligence (XAI).

METHODOLOGY

Dataset

Description

The Cotton_Disease dataset was utilized in this study. The dataset contains images categorized into four classes: diseased cotton leaf, diseased cotton plant, fresh cotton leaf, and fresh cotton plant. The dataset was divided into training, validation, and testing subsets to enable objective performance evaluation and to prevent data leakage between model development and assessment stages.

Data Pre-Processing

Prior to training, all images were preprocessed. For the baseline Convolutional Neural Network (CNN), images were resized to 128×128 pixels. For transfer-learning architectures such as ResNet, VGGNet, and EfficientNet, images were resized to 224×224 pixels, matching the predefined model input dimensions.

Pixel values were normalized to the range [0,1] to stabilize optimization. Class labels were converted to one-hot encoded vectors for multi-class classification. To improve generalization, data augmentation

techniques such as random rotation, zooming, shearing, and horizontal flipping were applied, simulating natural variability commonly observed in agricultural imagery.

Baseline CNN Architecture: A baseline CNN was implemented to establish an initial performance benchmark. The network consisted of sequential convolutional layers employing Rectified Linear Unit (ReLU) activations, followed by max-pooling layers to progressively downsample spatial dimensions. Dropout layers were incorporated to reduce overfitting. The extracted features were flattened and fed into dense layers, concluding with a Softmax output layer for four-class classification. This model served as the basis for comparison with more advanced pretrained architectures.

Transfer Learning Architectures

ResNet

The ResNet50 architecture, pretrained on ImageNet, was employed as a feature extractor (He et al., 2016). The fully connected classification layer was removed and replaced with a custom classifier consisting of Global Average Pooling, dense layers, dropout, and a Softmax output layer. Initially, all pretrained layers were frozen to exploit previously learned visual representations. Subsequently, selected deeper layers were unfrozen and fine-tuned using a reduced learning rate to adapt the model to cotton disease characteristics.

VGGNet

The VGG16 architecture was also evaluated (Simonyan & Zisserman, 2014). The original classification head was removed and replaced with a customized dense classifier incorporating dropout and Softmax activation. Training was performed in two stages: (a) training the new classifier while freezing convolutional layers and (b) fine-tuning selected upper layers to enhance feature specialization.

EfficientNet

EfficientNetB0 was selected due to its compound scaling strategy, which jointly optimizes network width, depth, and input resolution for improved performance and computational efficiency (Tan & Le, 2019). A pretrained EfficientNetB0 backbone, initialized with ImageNet weights, was retained as a feature extractor to leverage learned low- and mid-level visual representations. The original classification layers were replaced with a task-specific classification head tailored to the target classes. Fine-tuning was performed using a low learning rate to preserve the pretrained knowledge while allowing gradual adaptation to the agricultural imagery domain, thereby improving model generalization and classification accuracy.

Model Training Strategy

All models were trained using categorical cross-entropy loss and the Adam optimizer. Early stopping was applied to terminate training when validation loss ceased improving at a predefined patience level, thereby preventing overfitting. Model checkpointing was implemented to retain the best-performing weights, and learning-rate scheduling was used when performance plateaued.

Evaluation Metrics

Model performance was assessed using training accuracy, validation accuracy, test accuracy, and validation loss metrics. To improve interpretability, Explainable Artificial Intelligence (XAI) method such as Local Interpretable Model-Agnostic Explanations (LIME) was employed to visualize regions most influential in classification outcomes.

Model Comparison and Selection

The models were comparatively analyzed based on classification accuracy, generalization performance, susceptibility to overfitting or underfitting, and interpretability. The model demonstrating the most favorable balance across these criteria was selected as the optimal classifier for cotton disease recognition.

RESULTS

A. Overall Model Performance

All models were trained on the Cotton Disease dataset and evaluated on an independent test set. Table I summarizes the classification accuracy obtained for each model architecture.

TABLE I
COMPARISON OF MODEL ACCURACY

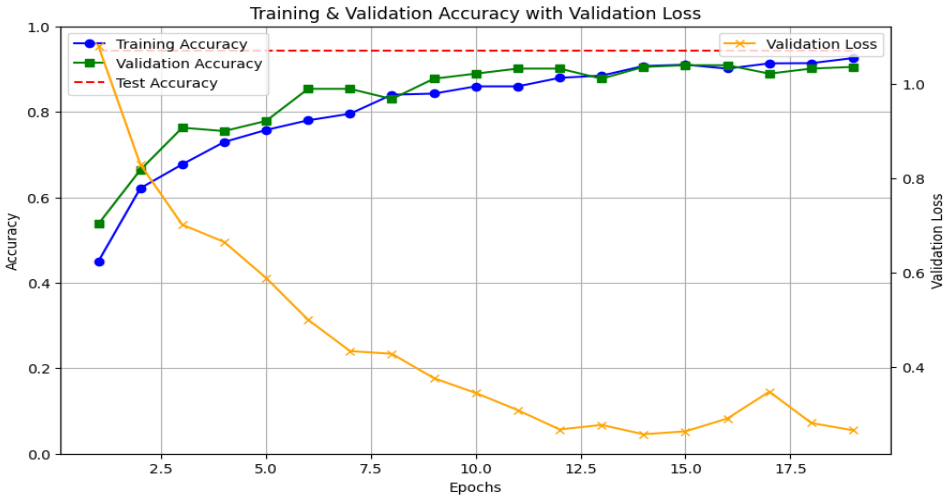
Model	Training Accuracy	Validation Accuracy	Test Accuracy
Baseline CNN	0.907	0.905	0.943
VGGNet	0.982	0.980	0.981
ResNet	0.959	0.964	0.962
EfficientNet	0.962	0.992	0.962

B. Loss and Accuracy Trends

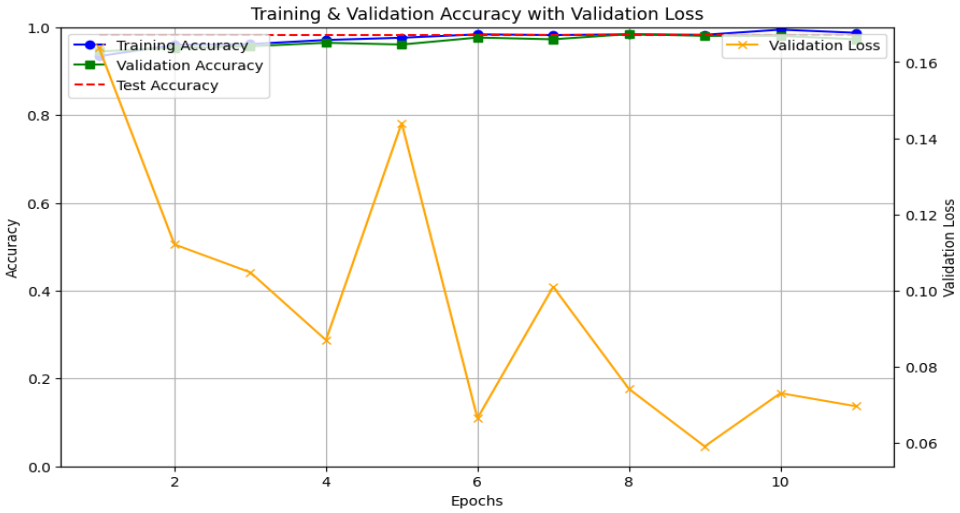
Below illustrates the learning curves for each model. The pretrained architectures converged faster and exhibited reduced overfitting relative to the baseline CNN, confirming the benefit of feature reuse from large-scale pretraining.

Figs. 1

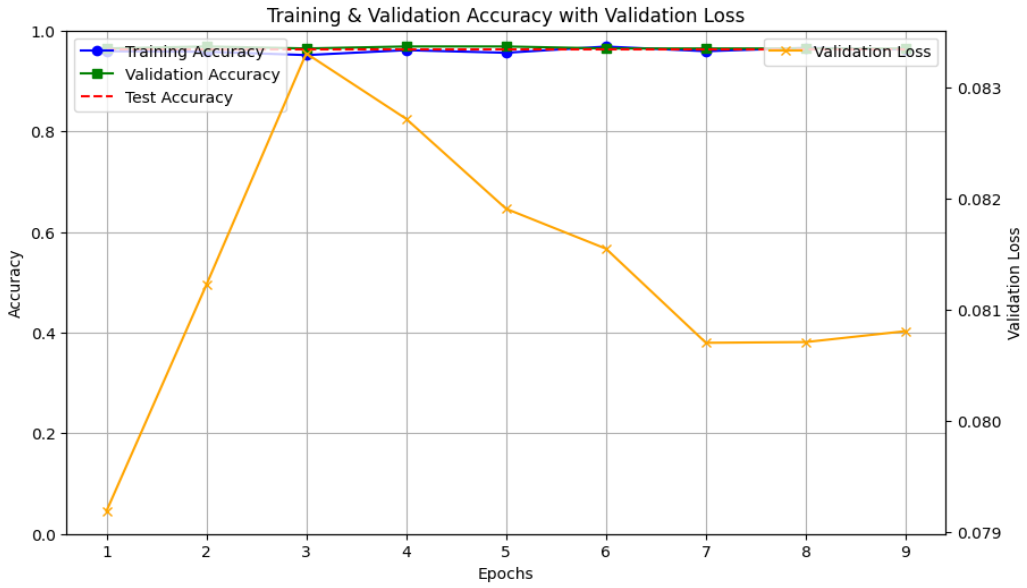
BASELINE CNN MODEL:



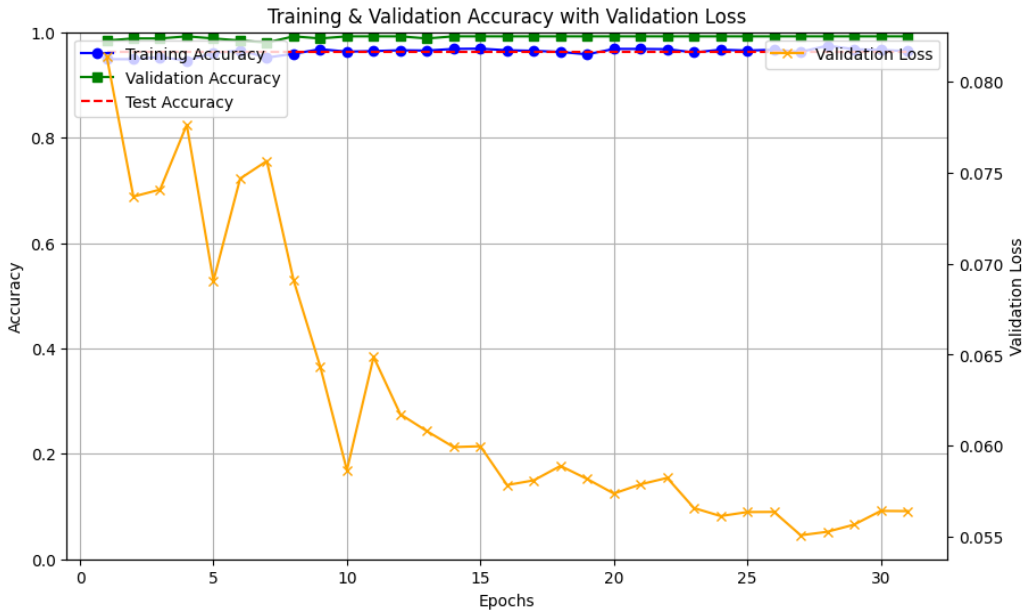
VGGNet MODEL:



ResNet MODEL:



EfficientNet MODEL:



C. Explainable AI (XAI) Interpretation

Explainable AI technique (LIME) was applied to visualize salient image regions, contributing to predictions. Superpixel-based explanation maps are shown below

Figs. 2

. *XAI visualization showing regions influencing classification decisions.*

Original Image
True: diseased cotton plant



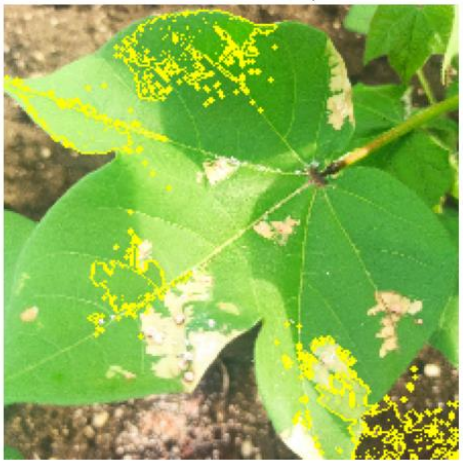
LIME Explanation
Prediction: fresh cotton plant

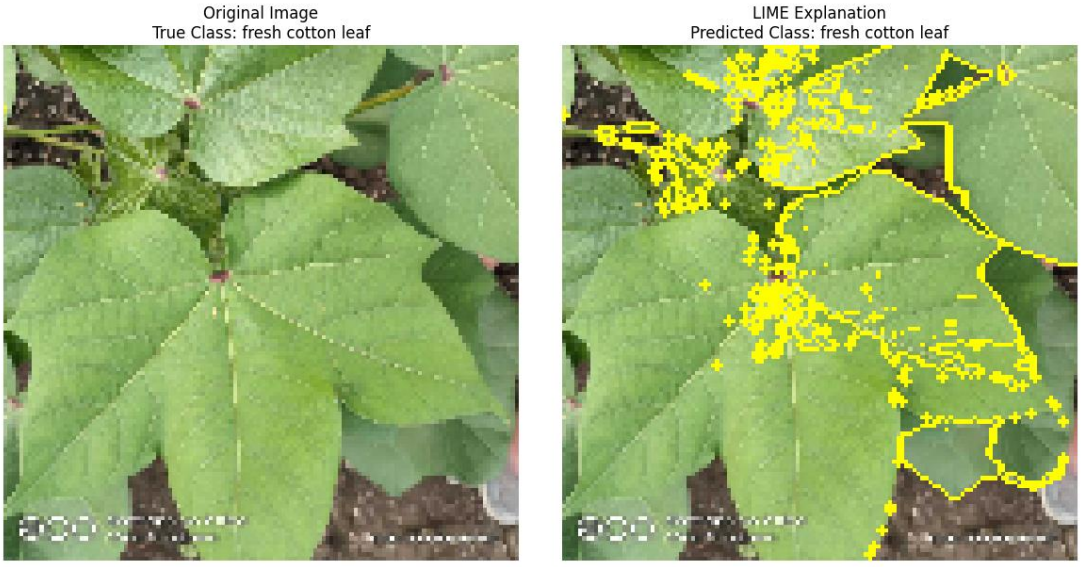


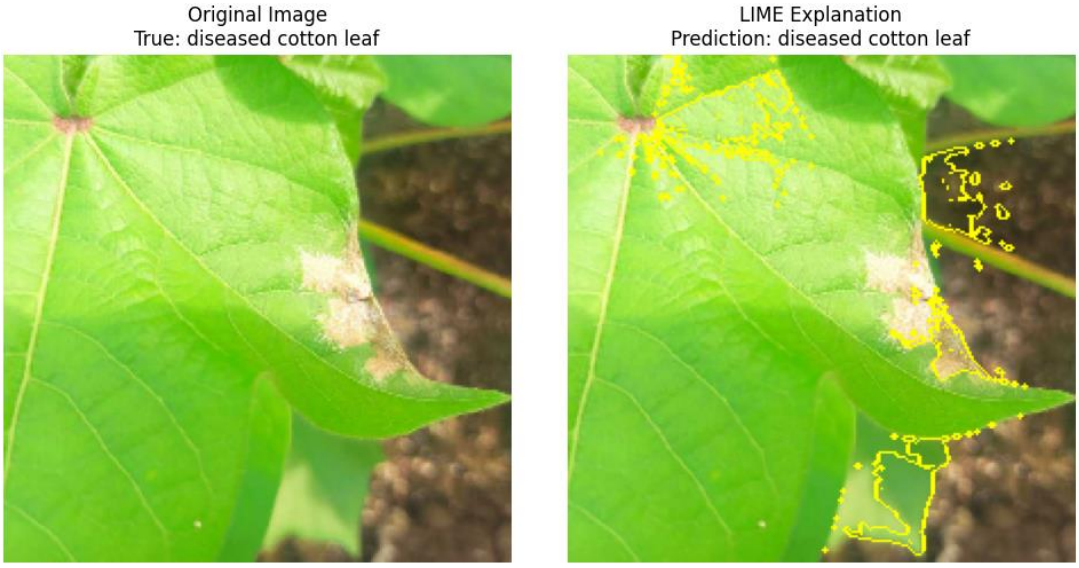
Original Image
True: diseased cotton leaf



LIME Explanation
Prediction: fresh cotton plant







The visualizations above show examples of both correct and incorrect classifications and also demonstrate that the models are primarily focused on **disease spots, discoloration regions, and texture variations**, confirming biologically meaningful reasoning rather than reliance on background artifacts.

CONCLUSION

This study evaluated multiple convolutional neural network architectures for automated classification of cotton leaf and plant diseases. A baseline CNN, VGGNet, ResNet, and EfficientNet were trained and tested on a curated Cotton_Disease dataset consisting of four classes: diseased cotton leaf, diseased cotton plant, fresh cotton leaf, and fresh cotton plant.

Experimental results demonstrated that transfer-learning–based models consistently outperformed the baseline CNN. Among the evaluated architectures, **EfficientNet achieved the highest validation accuracy (0.992)**, indicating strong generalization capability, while **VGGNet delivered the most balanced overall performance** across training, validation, and test sets. ResNet also performed competitively, showing high robustness with reduced underfitting relative to the baseline model. These results confirm that pretrained deep networks effectively leverage prior to learned representations to improve agricultural image classification tasks.

To further enhance the interpretability of the classification process, **LIME (Local Interpretable Model-Agnostic Explanations)** was integrated. LIME provided visual explanations highlighting the image regions most influential in model predictions. This capability is particularly valuable in agricultural decision-support systems, as it promotes transparency, facilitates trust among domain experts, and enables diagnostic validation when misclassifications occur.

Overall, the findings indicate that pretrained CNN architectures are highly suitable for plant-disease recognition, with EfficientNet and VGGNet showing the strongest predictive capability. Moreover, the integration of explainable AI tools such as LIME contributes to more interpretable and trustworthy deep-learning systems. Future work may extend this framework by incorporating larger and more diverse datasets, evaluating additional state-of-the-art architectures, and exploring real-time deployment in field environments.